

PIERRE WILMOT

MACHINE LEARNING ENGINEER · FOUNDER, THE KLONE FACTORY

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Software engineer with 10+ years shipping production systems, largely focused on deep learning for visual content — neural rendering, 3D reconstruction and generative pipelines — at PlayStation, Dimension and Samsung Research. Comfortable from low-level C++/CUDA optimisation up to architecture and research direction, and equally at home taking a product to market: I founded The Klone Factory, building the hardware, the capture and rendering pipeline and the commercial side end to end.

DEEP LEARNING

PyTorch, neural rendering, Gaussian splatting, NeRF, diffusion (Stable Diffusion, LoRA, ControlNet), segmentation, depth, keypoints

DOMAINS

Virtual production, facial animation rigs (Maya), photogrammetry, AR/VR, video games, low-latency trading systems

ENGINEERING

C, C++, CUDA, OpenCL, Python, Lua, real-time optimisation, build systems, web-native 3D viewers

HARDWARE

Electronics, PCB design, control systems, CAD, 3D printing, machining

EXPERIENCE

The Klone Factory — Founder

2026 — Present

3D digitisation service for auction houses and antique dealers: remote bidders working from flat catalogue photos can't judge condition, so they bid less or not at all. Built and run the whole company solo.

- **Hardware** — designed and built the capture rig from scratch (electronics, control systems, PCB design) for repeatable, high-quality capture inside a working auction house rather than a lab.
- **Software** — full pipeline: photogrammetry and Gaussian splatting reconstruction, a fast web-native 3D viewer, and provenance/condition notes anchored to the model.
- **Commercial** — prospecting and closing auction house clients, client relationships, marketing and content, pricing and strategy.

PlayStation (Sony Interactive Entertainment) — Machine Learning Engineer

2025 — 2026

Deep learning pipeline for high-fidelity facial animation: animating production-ready Maya face rigs directly from head-mounted camera footage. Built and optimised the neural architectures mapping video to rig control parameters from a small set of artist-annotated sequences, cutting manual animation work on cinematic game content.

Dimension — Senior Engineer (software & ML lead)

2022 — 2025

Owned software and ML strategy for a virtual production studio (the CTO owned camera, volume and LED-wall hardware). Worked directly with producers, VFX artists and on-set crew to find where ML could move quality and cost, then built it.

- Moved asset capture from mesh-and-texture pipelines to NeRF and Gaussian splatting, keeping view-dependent effects (reflections, subsurface scattering, speculars) that read as lifelike on screen — shifting splats from background capture to **hero assets**.
- Built material acquisition via differential rendering for relighting, scene understanding (segmentation, depth) for virtual production, and generative image/video pipelines.

Mirriad — Senior C++ and Machine Learning Engineer

2021

Developed and integrated PyTorch models for automatic colour grading and keypoint detection in the company's compositing software, plus live player and ball tracking to embed advertising into sports broadcasts in real time.

Samsung Research UK — Machine Learning Engineer, Extended Reality

2019 — 2021

Meta-learning approach for generating textile textures for virtual avatar clothing; core contributor to the avatar-from-capture-rig pipeline (3D projection, segmentation, hair reconstruction); ported OpenPose to Android with multi-person face and body keypoint estimation. Two internal awards for software architecture and code review practice.

Fetch.ai — Machine Learning Engineer

2019

Designed, implemented and optimised the machine learning framework for the Fetch virtual machine, including a C++ compute-graph implementation powering a semantic search engine. Open sourced under Apache 2.0.

Deutsche Börse — Software Engineer, EUREX exchange

2018

Became de facto technical lead on the exchange codebase and led development of the TRF Baskets feature. Decoupled application logic from messaging and database layers so the system could be tested and run outside the full ecosystem — a more robust codebase and a markedly faster development cycle for a team new to C++.

Artomatix — Deep Learning Engineer / Researcher

2016 — 2018

State-of-the-art results in parametric neural texture synthesis and style transfer, published as *Stable and Controllable Neural Texture Synthesis and Style Transfer Using Histogram Losses*. Also neural texture weathering, texture segmentation and single-image super resolution, and a neural network runtime written from scratch in C++/OpenCL and shipped in the product.

Earlier

- **Ubisoft**, Software Engineer (2015) — built the mobile build and unit-test system for iOS and Android; contributed to the cross-platform Ubiservices online-features library.
- **Blackmagic Design**, Infrastructure Engineer (2014) — one-click Python build system with a web UI: checkout, multi-platform compile, test, ship-ready packaging.
- **Good Game Productions**, Video Game Programmer (2014) — ported the in-house game framework from Cocos2d (Objective-C) to Cocos2d-x (C++), giving designers a drag-and-drop scene editor.
- **OutOfTheBit**, Junior Developer (2013) — shipped three mobile games in a year across iOS, Android and Windows Phone; Four in a Row and Dots & Boxes reached #1 and #2 in the App Store charts.

SELECTED WORK & PUBLICATIONS**The Klone Factory**

3D capture studio for auction houses and dealers — hardware, pipeline and viewer.

Stable and Controllable Neural Texture Synthesis and Style Transfer Using Histogram Losses

Wilmot, Risser, Barnes — 2017

Shipped games

2048, Snake, Laser Maze, Four In A Row, Dots & Boxes, SoundRide — published on iOS and Android.

EDUCATION

Epitech — BSc Computer Science
2010 — 2012

Collège de France — Deep Learning course, Yann LeCun
2016